

Lab Report Guidelines

Lab 4: Incremental Pattern Mining & Correlation on YouTube Comments

1. Cover Page

2. Objective

Briefly describe what the lab aims to achieve.

3. Tools and Libraries Used

List all software and Python libraries.

4. Dataset Description

Dataset Used: `cleaned_comments.csv` (from Lab 2)

- Contains YouTube comments with preprocessed token lists (`cleaned_tokens`).
- Each comment represents a transaction of words.
- The data is segmented into 5 chunks to simulate the arrival of comments over time.

5. Methodology / Procedure

5.1 Data Loading and Segmentation

- Load `cleaned_comments.csv` using pandas.
- Ensure `cleaned_tokens` column exists and convert to lists if needed.
- Split the dataset into **five equal chunks** using `numpy.array_split()`.

5.2 Pattern Mining per Chunk

- For each chunk:
 - Extract tokens as transactions.
 - Count top 10 **unigrams** (individual words).
 - Count top 10 **co-occurring word pairs** using `itertools.combinations()`.
 - Visualize frequency distributions using bar charts.

5.3 Tracking Pattern Evolution

- Record the frequency of top 5–10 unigrams and pairs across all chunks.
- Create a table to track how their frequencies change over time.

5.4 Correlation Analysis

- Select 3–5 interesting patterns (e.g., “exam”–“stress”).
- Calculate their frequencies across the 5 chunks.
- Build a correlation matrix using `pandas.corr()`.
- Plot frequency trends and interpret correlations.

5.5 Reflection and Insight

- Write short explanations for each pattern:
 - How its frequency changes.
 - Any correlations with other words.
 - Possible contextual reasons for these changes.

6. Code Implementation

Include well-commented code for:

- Data loading and splitting

- Pattern counting (unigrams & pairs)
- Visualization (matplotlib plots)
- Correlation computation

7. Results and Discussion

Include tables, plots, and insights such as:

- Top 10 words and pairs per chunk
- Frequency-over-time graphs
- Correlation matrix

Discuss:

- Which patterns appeared or disappeared over time.
- How correlations reveal relationships between topics.
- Challenges faced during analysis.

8. Conclusion

Summarize your findings.

Submission Requirements:

Please upload the following:

1. Lab Report (PDF format)
2. Python Code File (.py or .ipynb)

Deadline: Submit before the due date.